

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 9: Singular Value Decomposition (SVD)

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1 Keywords

- Singular Value Decomposition, SVD, spectral norm, condition number, general linear maps, ridge regression

2 Main Points

2.1 Singular Value Decomposition

- Any $m \times p$ matrix, $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ (for a design matrix, m is the number of data points and p the number of features)
 - $\mathbf{U}$ is an $m \times m$ orthogonal matrix
 - $\mathbf{S}$ is an $m \times p$ matrix with zeros everywhere except the diagonal where $S_{ii} = s_i \geq 0$
 - $\mathbf{V}$ is a $p \times p$ orthogonal matrix
- The values s_i are known as the *singular values* of $\mathbf{X}$
- The columns $\mathbf{u}_i$ of $\mathbf{U}$ and $\mathbf{v}_i$ of $\mathbf{V}$ are the left and right *singular vectors*. Column by column the SVD says $\mathbf{X} \mathbf{v}_i = s_i \mathbf{u}_i$ and $\mathbf{X}^T \mathbf{u}_i = s_i \mathbf{v}_i$
- If $\mathbf{X}$ is symmetric and positive semi-definite (e.g. a covariance matrix) then $\mathbf{U} = \mathbf{V}$ and $\mathbf{S} = \mathbf{\Lambda}$: the SVD is the eigen-decomposition. For a general symmetric matrix $s_i = |\lambda_i|$, and the columns of $\mathbf{U}$ and $\mathbf{V}$ belonging to a negative eigenvalue differ in sign
- **Where it comes from.** The $(m+p) \times (m+p)$ matrix $\mathbf{B} = \begin{pmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{0} \end{pmatrix}$ is symmetric. Writing an eigenvector as $(\mathbf{u}, \mathbf{v})$, the equation $\mathbf{B}(\mathbf{u}, \mathbf{v}) = s(\mathbf{u}, \mathbf{v})$ says exactly $\mathbf{X} \mathbf{v} = s \mathbf{u}$ and $\mathbf{X}^T \mathbf{u} = s \mathbf{v}$. So everything we know about symmetric matrices (real eigenvalues, orthogonal eigenvectors) carries over to every matrix. The lecture works through the details, including why $\mathbf{u}$ and $\mathbf{v}$ each need rescaling by $\sqrt{2}$
- **Economical SVD**
 - If $m > p$ some algorithms won't bother outputting the last $m - p$ columns of $\mathbf{U}$
 - If $p > m$ some algorithms won't bother outputting the last $p - m$ columns of $\mathbf{V}$
 - In this case it will output a square $\mathbf{S}$ matrix
 - In numpy, `U, s, Vt = np.linalg.svd(X)` gives the full form and `np.linalg.svd(X, full_matrices=False)` the economical one. Note that numpy returns the singular values as a vector `s` and returns $\mathbf{V}^T$, not $\mathbf{V}$
- **Finding the SVD.** In principle we can take whichever of $\mathbf{X} \mathbf{X}^T$ and $\mathbf{X}^T \mathbf{X}$ is smaller, find its eigenvalues s_i^2 and eigenvectors, and recover the other set of singular vectors from $\mathbf{U} = \mathbf{X} \mathbf{V} \mathbf{S}^{-1}$ (for the non-zero s_i). Good libraries do not do this: forming $\mathbf{X}^T \mathbf{X}$ squares the condition number (see the mappings notes), so they use a more stable algorithm on $\mathbf{X}$ itself

2.2 Spectrum and the Spectral Norm

- The singular values, ordered $s_1 \geq s_2 \geq \dots \geq 0$, are the (singular value) *spectrum* of $\mathbf{X}$
- The largest, $s_{\max}$, is the most that $\mathbf{X}$ can stretch a vector. It is the *spectral norm* from the vector spaces lecture:

$$\|\mathbf{X}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{X}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = s_{\max}$$

- **Proof.** Write $\mathbf{x} = \sum_i c_i \mathbf{v}_i$ in the orthonormal basis of right singular vectors, so $\|\mathbf{x}\|^2 = \sum_i c_i^2$. Then $\mathbf{X}\mathbf{x} = \sum_i c_i s_i \mathbf{u}_i$, and as the $\mathbf{u}_i$ are orthonormal too

$$\|\mathbf{X}\mathbf{x}\|^2 = \sum_i s_i^2 c_i^2 \leq s_{\max}^2 \sum_i c_i^2 = s_{\max}^2 \|\mathbf{x}\|^2$$

with equality when $\mathbf{x} = \mathbf{v}_{\max}$, which is mapped to $\mathbf{X}\mathbf{v}_{\max} = s_{\max} \mathbf{u}_{\max}$

- Consequences
 - For a symmetric matrix $\|\mathbf{X}\|_2 = \max_i |\lambda_i|$, as used in the eigensystems notes
 - For any square matrix $\|\mathbf{X}\|_2 \geq \max_i |\lambda_i|$ (if $\mathbf{X}\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v}$ is stretched by $|\lambda|$, which cannot exceed $s_{\max}$). The two can differ a lot for non-symmetric matrices (Exercise 3)
 - $\|\mathbf{X}\|_F^2 = \sum_{ij} X_{ij}^2 = \sum_i s_i^2 \geq s_{\max}^2$, so $\|\mathbf{X}\|_2 \leq \|\mathbf{X}\|_F$ (Exercise 2)
 - For a square invertible matrix the smallest stretch is $s_{\min}$ and $\|\mathbf{X}^{-1}\|_2 = 1/s_{\min}$, so the condition number is $\kappa = \|\mathbf{X}\|_2 \|\mathbf{X}^{-1}\|_2 = s_{\max}/s_{\min}$. This generalises the eigenvalue ratio for symmetric matrices
 - Spectral normalisation of a network layer (vector spaces notes) divides $\mathbf{L}$ by $s_{\max}$ so that $\|\mathbf{L}\|_2 = 1$

2.3 General Linear Mapping

- Recall that matrices are the most general linear operators
- Since any matrix $\mathbf{X}$ can be written as $\mathbf{U}\mathbf{S}\mathbf{V}^T$ we can interpret any linear mapping as doing three operations
 - A rotation (with possibly a reflection) defined by $\mathbf{V}^T$
 - A rescaling of each coordinate by s_i
 - A rotation (with possibly a reflection) defined by $\mathbf{U}$
- When $\mathbf{X}$ is not square, $\mathbf{S}$ also either squashes some directions to zero or embeds the result in a space with extra dimensions
- The *rank* of any matrix is its number of non-zero singular values (the number of linearly independent rows or columns)
- For a square matrix, using $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ and $|\mathbf{U}| = \pm 1$ for an orthogonal matrix, $|\det \mathbf{X}| = \prod_i s_i$: the singular values multiply up to the change of volume
- **Duality**
 - Using $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ then
 - $\mathbf{C} = \mathbf{X}\mathbf{X}^T = \mathbf{U}\mathbf{S}\mathbf{S}^T\mathbf{U}^T$
 - $\mathbf{D} = \mathbf{X}^T\mathbf{X} = \mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T$
 - $\mathbf{S}\mathbf{S}^T$ and $\mathbf{S}^T\mathbf{S}$ are diagonal matrices whose non-zero diagonal elements are s_i^2

2.4 Ridge Regression

- Ridge regression is linear regression with an L_2 regulariser
- Adding a regulariser $\eta \|\mathbf{w}\|^2$ to the squared loss, the weights, $\mathbf{w}^*$, that minimise the loss function are given by $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$
- Using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then

$$\mathbf{w}^* = \mathbf{V} \hat{\mathbf{S}}^+ \mathbf{U}^T \mathbf{y}$$

where $\hat{\mathbf{S}}^+$ is a regularised pseudo-inverse of $\mathbf{S}$ given by

$$\hat{\mathbf{S}}^+ = (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^T$$

- If $\eta = 0$ (and all $s_i > 0$) this is equal to the pseudo-inverse of $\mathbf{S}$
- $\hat{\mathbf{S}}^+$ is a $p \times m$ matrix which is zero everywhere except on the diagonal, where $\hat{S}_{ii}^+ = \frac{s_i}{s_i^2 + \eta}$
 - Note if $s_i = 0$ linear regression has an infinity of solutions and the pseudo-inverse of $\mathbf{X}$, as we derived it, does not exist (setting $\eta = 0$ we get $S_{ii}^+ = 1/s_i$ which is not defined when $s_i = 0$). See the Moore–Penrose pseudo-inverse below
 - In the regularised case $\hat{S}_{ii}^+ = 0$ (we have selected one of the solutions that minimise the squared error)
 - If $s_i \ll \eta$ then without the regularisation term the inverse is very ill-conditioned while with the regularisation term $\hat{S}_{ii}^+$ will be small
 - If $s_i \gg \eta$ then $\hat{S}_{ii}^+ \approx \frac{1}{s_i} = S_{ii}^+$
- Adding an L_2 regulariser means that the optimum weights, $\mathbf{w}^*$, will be less sensitive to the training data reducing the variance in the bias-variance dilemma

2.5 The Moore–Penrose Pseudo-Inverse (going further)

- The pseudo-inverse we derived, $\mathbf{X}^+ = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$, only exists when $\mathbf{X}^T \mathbf{X}$ is invertible. That needs every one of the p singular values to be non-zero, which is impossible when $m < p$ (the under-constrained case)
- Using the SVD it can be written $\mathbf{X}^+ = \mathbf{V} \mathbf{S}^+ \mathbf{U}^T$, where $\mathbf{S}^+$ is the $p \times m$ diagonal matrix with $S_{ii}^+ = 1/s_i$
- The *Moore–Penrose pseudo-inverse* makes one change: wherever $s_i = 0$ it sets $S_{ii}^+ = 0$ instead of $1/s_i$

$$S_{ii}^+ = \begin{cases} 1/s_i & s_i \neq 0 \\ 0 & s_i = 0 \end{cases}$$

This always exists. When $\mathbf{X}^T \mathbf{X}$ is invertible it is the same as our pseudo-inverse, and when $\mathbf{X}$ is square and invertible it is $\mathbf{X}^{-1}$

- $\mathbf{w} = \mathbf{X}^+ \mathbf{y}$ minimises $\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2$ and, of all the weight vectors that do so, it is the one with the smallest $\|\mathbf{w}\|$
- To see why, rotate to the coordinates given by the SVD: let $\mathbf{z} = \mathbf{V}^T \mathbf{w}$ and $\mathbf{c} = \mathbf{U}^T \mathbf{y}$. Orthogonal matrices do not change lengths, so $\|\mathbf{w}\| = \|\mathbf{z}\|$ and

$$\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2 = \|\mathbf{U}^T (\mathbf{U} \mathbf{S} \mathbf{V}^T \mathbf{w} - \mathbf{y})\|^2 = \|\mathbf{S} \mathbf{z} - \mathbf{c}\|^2 = \sum_{i: s_i \neq 0} (s_i z_i - c_i)^2 + \sum_{i: s_i = 0} c_i^2$$

(in the second sum i runs over the rows of $\mathbf{S}$ that have no non-zero singular value)

- The loss is minimised by $z_i = c_i/s_i$ whenever $s_i \neq 0$
- The other components of z do not appear in the loss at all: these are the directions v_i with $\mathbf{X} v_i = \mathbf{0}$, where the data give no information
- Every choice of them fits equally well. Setting them to zero gives the smallest $\|z\| = \|w\|$, and that is exactly what $S_{ii}^+ = 0$ does
- It is also the limit of ridge regression as the regularisation vanishes: as $\eta \rightarrow 0$, $\hat{S}_{ii}^+ = s_i/(s_i^2 + \eta)$ tends to $1/s_i$ if $s_i \neq 0$ and is 0 if $s_i = 0$
- This is what `np.linalg.pinv` computes, and `np.linalg.lstsq` returns the same minimum-norm solution. In practice "zero" means below a small tolerance, so tiny singular values caused by rounding errors are discarded rather than inverted

3 Exercises

3.1 Ridge regression

- Ridge regression is just linear regression with an L_2 regulariser
 1. Derive the optimal weights in ridge regression
 2. Show that using $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ then $w^* = \mathbf{V} (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^T \mathbf{U}^T y$
- See answers

3.2 Spectral norm

1. Show that $\|\mathbf{X} x\|_2 \leq s_{\max} \|x\|_2$, with equality for $x = v_{\max}$
2. Show that $\|\mathbf{X}\|_F^2 = \text{Tr } \mathbf{X}^T \mathbf{X} = \sum_i s_i^2$, and hence that $\|\mathbf{X}\|_2 \leq \|\mathbf{X}\|_F$
3. For $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
 - (a) Show that both eigenvalues of $\mathbf{J}$ are 1
 - (b) Find the singular values of $\mathbf{J}$ from the eigenvalues of $\mathbf{J}^T \mathbf{J}$, and hence $\|\mathbf{J}\|_2$ and the condition number
 - (c) Repeat for the symmetric matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, whose eigenvalues are 3 and -1

4 Experiments

4.1 SVD

- Check the claims in these notes on a random 3×4 matrix

```
import numpy as np

rng = np.random.default_rng(0)
X = rng.normal(size=(3, 4))
U, s, Vt = np.linalg.svd(X)      # s is a vector, Vt is V transposed
S = np.zeros_like(X)
np.fill_diagonal(S, s)           # the 3 x 4 matrix S
V = Vt.T
print(np.allclose(U @ S @ Vt, X))      # X = U S V^T
print(np.allclose(U @ U.T, np.eye(3)), np.allclose(V.T @ V, np.eye(4)))
```

```

print(np.linalg.eigvalsh(X @ X.T)[:,-1], s**2)          # eigenvalues are s_i^2
print(np.linalg.eigvalsh(X.T @ X)[:,-1])              # the same, plus a 0

# spectral norm: the largest stretch of any unit vector
print(np.linalg.norm(X, 2), s[0])                     # both s_max
print(np.linalg.norm(X @ V[:, 0]), np.linalg.norm(X @ V[:, 1])) # s_1, s_2
x = rng.normal(size=4)
print(np.linalg.norm(X @ x) <= s[0] * np.linalg.norm(x)) # never above s_max
print(np.linalg.norm(X, 'fro')**2, np.sum(s**2))        # Frobenius^2

# ridge regression through the SVD
eta = 0.1
lhs = np.linalg.inv(X.T @ X + eta * np.eye(4))
rhs = V @ np.linalg.inv(S.T @ S + eta * np.eye(4)) @ V.T
print(np.allclose(lhs, rhs))
print(np.linalg.cond(X.T @ X))    # X^T X is singular (rank 3): enormous

```

- Every check prints True. The unit vector v_1 is stretched by exactly s_1 and v_2 by s_2
- $\mathbf{X}^T \mathbf{X}$ is 4×4 but only has rank 3, so it has no inverse. Rounding errors mean numpy may still return one, with a condition number around 10^{16} : a warning sign never to ignore. Adding $\eta \mathbf{I}$ makes it safely invertible

5 Answers

5.1 Ridge regression

1. It is straightforward to show

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

2. The only hard part is to show that

$$(\mathbf{X}^T \mathbf{X} + \eta \mathbf{I})^{-1} = \mathbf{V} (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{V}^T$$

- It is easy to show that $\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T$
- But we also have $\mathbf{I} = \mathbf{V} \mathbf{V}^T$ as $\mathbf{V}$ is an orthogonal matrix
- Thus $\mathbf{M} = \mathbf{X}^T \mathbf{X} + \eta \mathbf{I} = \mathbf{V} (\mathbf{S}^T \mathbf{S} + \eta \mathbf{I}) \mathbf{V}^T = \mathbf{V} \mathbf{W} \mathbf{V}^T$ where $\mathbf{W} = \mathbf{S}^T \mathbf{S} + \eta \mathbf{I}$
- But $(\mathbf{A} \mathbf{B} \mathbf{C})^{-1} = \mathbf{C}^{-1} \mathbf{B}^{-1} \mathbf{A}^{-1}$ (which we can verify by multiplying $\mathbf{C}^{-1} \mathbf{B}^{-1} \mathbf{A}^{-1}$ on either the left or right by $\mathbf{A} \mathbf{B} \mathbf{C}$)
- Thus $\mathbf{M}^{-1} = (\mathbf{V} \mathbf{W} \mathbf{V}^T)^{-1} = (\mathbf{V}^T)^{-1} \mathbf{W}^{-1} \mathbf{V}^{-1} = \mathbf{V} \mathbf{W}^{-1} \mathbf{V}^T$ where we use $\mathbf{V}^{-1} = \mathbf{V}^T$ as $\mathbf{V}$ is an orthogonal matrix

5.2 Spectral norm

1. See the proof in the main points: expand $\mathbf{x}$ in the right singular vectors, so $\|\mathbf{X} \mathbf{x}\|^2 = \sum_i s_i^2 c_i^2 \leq s_{\max}^2 \sum_i c_i^2$. For $\mathbf{x} = \mathbf{v}_{\max}$ only one c_i is non-zero and the inequality is an equality
2. $\sum_{ij} X_{ij}^2 = \sum_j (\mathbf{X}^T \mathbf{X})_{jj} = \text{Tr } \mathbf{X}^T \mathbf{X}$. With $\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T$ and the cyclic property of the trace this is $\text{Tr } \mathbf{S}^T \mathbf{S} = \sum_i s_i^2$. Every term is non-negative, so the sum is at least $s_{\max}^2$
3. Two 2×2 matrices

- (a) $\mathbf{J}$ is triangular, so its eigenvalues are its diagonal entries, 1 and 1

- (b) $\mathbf{J}^T \mathbf{J} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has $\lambda^2 - 3\lambda + 1 = 0$, so $\lambda = (3 \pm \sqrt{5})/2$ and the singular values are $s = (\sqrt{5} \pm 1)/2$, i.e. 1.618 and 0.618. So $\|\mathbf{J}\|_2 = (1 + \sqrt{5})/2 \approx 1.618$ (the golden ratio) although every eigenvalue is 1, and $\kappa = 1.618/0.618 \approx 2.62$. The eigenvalues of a non-symmetric matrix do not tell you how much it stretches
- (c) Here $s_i = |\lambda_i| = 3$ and 1, so the spectral norm is 3 and $\kappa = 3$. For a symmetric matrix the eigenvalues do tell the whole story